ACME METALS

Digital Material Passport

ID **DMP-METAL-002** - *Version* **1.0.0**
Issue Date **2025-05-14** - *Certificate Type* **EN 10204 3.1**

Manufacturer	Customer	Business Transaction
ACME Metal Works GmbH Industrial Park 123 52066 Aachen DE quality@acme-metal.example.com	**Engineering Solutions Ltd.** Tech Park Way 45 Cardiff CF14 5DU GB procurement@engisolutions.example.com	**Order** PO-78902 · **2000 kg** Pos. 10 · 2025-04-21 **Delivery** DN-56790 · **2000 kg** Pos. 1 · 2025-05-13
Subcustomer **American Heavy Industries Inc.** 5000 Industrial Blvd Building C Detroit, MI 48201 US materials@heavyind.example.com	**Goods Receiver** **Canadian Construction Corp** 777 Construction Ave Toronto, ON M5H 2N2 CA logistics@canconstruct.example.ca	

Product

Product Name	Structural Steel S420N	*Name (EN)*	1.8902
Batch ID	H-10988-01	*Shape*	Plate - Width 2000 mm - Thickness 25 mm - Length 6000 mm
Surface Condition	Hot-rolled	*Product Norm*	EN 10025-3 (2019)
Weight	2000 kg		
Production Date	2025-05-10		
Country of Origin	DE		

Heat Treatment

Process: **Normalizing** - *Lot:* NORM-2024-0823-A12 - *Furnace:* NF-LINE-02 - *Date:* 2024-08-23

Stage	*Temperature*	*Duration*	*Cooling*	*Atmosphere*
Austenitizing	920 C	60 min		Air
Cooling	20 C		Air	

Chemical Analysis

Heat Number **H-10988** - *Melting Process* **EAF+LF+VD** - *Casting Method* **ContinuousCasting** - *Casting Date* **2025-05-09** - *Sample Location* **Ladle**

Symbol	C	Mn	Si	P	S	CEV
Unit	%	%	%	%	%	%
Min						
Max	0.20	1.60	0.50	0.025	0.015	0.44
Actual	0.16	1.48	0.28	0.016	0.010	0.41

CEV = C+Mn/6+(Cr+Mo+V)/5+(Ni+Cu)/15: 0.41%

Mechanical Properties

Property	*Symbol*	*Actual*	*Minimum*	*Maximum*	*Method*	*Status*

Tensile Strength — EN ISO 6892-1 ✓
Test temperature: 20°C
Specimen: 1/3R, L

No.	1	2	3	*Average*	*Min*	*Max*	*Std Dev*	*Spec Min*	*Spec Max*
Value [MPa]	558	562	560	560	558	562	2	520	680

Yield Strength — EN ISO 6892-1 ✓
Test temperature: 20°C, 3 specimens tested

No.	1	2	3	*Average*	*Min*	*Max*	*Std Dev*	*Spec Min*
Value [MPa]	442	445	448	445	442	448	3	420

Elongation after fracture — EN ISO 6892-1 ✓
Gauge length 5.65#S_0, 3 specimens tested

No.	1	2	3	*Average*	*Min*	*Max*	*Std Dev*	*Spec Min*
Value [%]	23	24	25	24	23	25	1	19

Reduction of Area[1] — EN ISO 6892-1 ✓

No.	1	2	3	*Average*	*Min*	*Max*	*Std Dev*	*Spec Min*
Value [%]	60	62	64	62	60	64	2	50

Charpy V-notch Impact Energy[2] EN ISO 148-1 ✓
Specimen: Top, T-L

No.	1	2	3	*Average*	*Min*	*Max*	*Std Dev*	*Spec Min*
Value [J]	56	58	60	58	56	60	2	40

Charpy V-notch Impact Energy[2] EN ISO 148-1 ✓
Specimen: Bottom, T-L

No.	1	2	3	*Average*	*Min*	*Max*	*Std Dev*	*Spec Min*
Value [J]	54	57	59	56.7	54	59	2.5	40

Brinell Hardness EN ISO 6506-1 ✓
Ball: 10mm, Force: 3000kg, 5 indentations

No.	1	2	3	4	5	*Average*	*Min*	*Max*	*Std Dev*	*Spec Min*	*Spec Max*
Value [HBW]	183	185	187	184	186	185	183	187	1.58	150	220

Vickers Hardness EN ISO 6507-1 ✓
10 kg load, 5 indentations

No.	1	2	3	4	5	*Average*	*Min*	*Max*	*Std Dev*	*Spec Min*	*Spec Max*
Value [HV10]	192	195	198	194	196	195	192	198	2.35	160	230

Rockwell Hardness	HR	18 HRC		22 HRC	EN ISO 6508-1	✓
Elastic Modulus	E	210 GPa			EN ISO 6892-1	✓
Strain Hardening Exponent	n	0.18			ASTM E646	✓
Plastic Strain Ratio	r	1.2	1.0		EN ISO 10113	✓

0.2% Proof Strength[1] EN ISO 6892-1 ✓

No.	1	2	3	*Average*	*Min*	*Max*	*Std Dev*	*Spec Min*
Value [MPa]	428	430	432	430	428	432	2	400

[1] 3 specimens tested [2] Test temperature: -20°C

Validation

We hereby certify that the material described above has been manufactured and tested in accordance with the requirements of EN 10204:2004 type 3.1 and the specified standards. The results comply with the requirements.

Validated by **Johann Weber**, Quality Inspector, Quality Assurance - 2025-05-14